Efficient Distributed Topology Control in 3-Dimensional Wireless Networks

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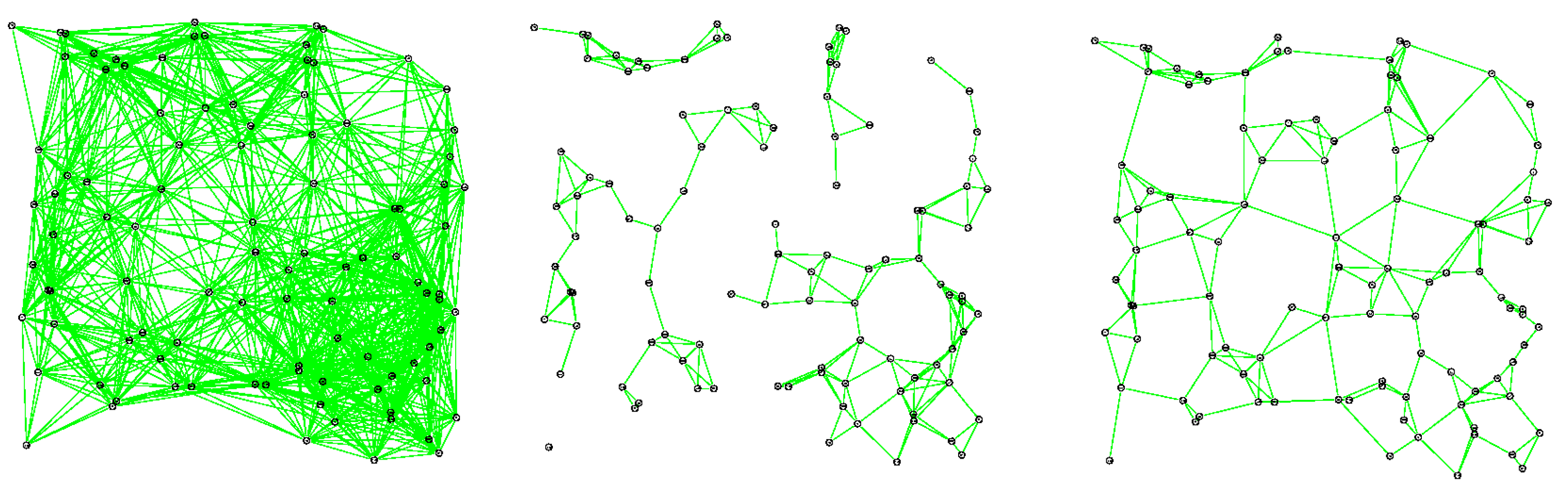
Motivation and 3-Dimensional Networks

Examples

No topology control: nodes transmit at maximum and minimum power

With topology control



Energy inefficient, high interference, low throughput, partitioned network

Challenges in 3-Dimensions

- No natural ordering of nodes based on angular information
- Very high node density to ensure connectivity
- Many 2-D algorithms not readily extensible to 3-D
- Constraints, e.g. structural
- High complexity

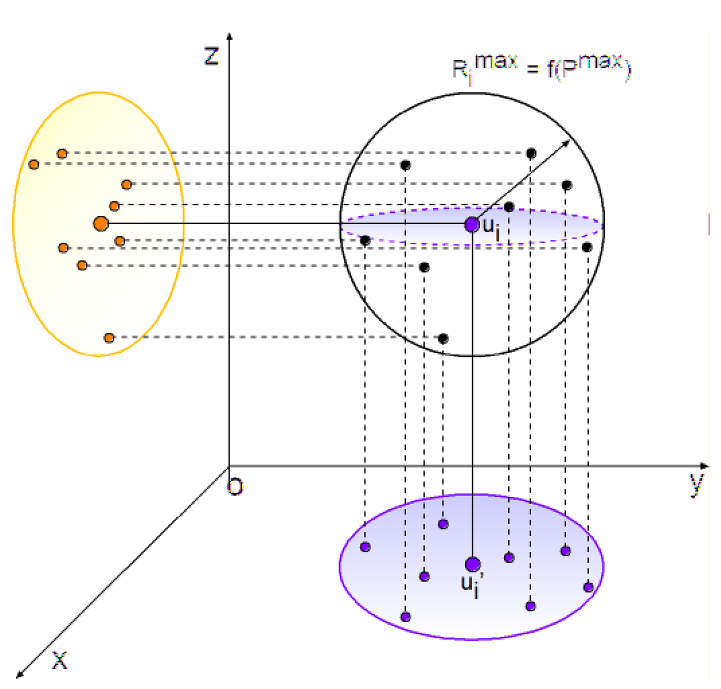
Goal

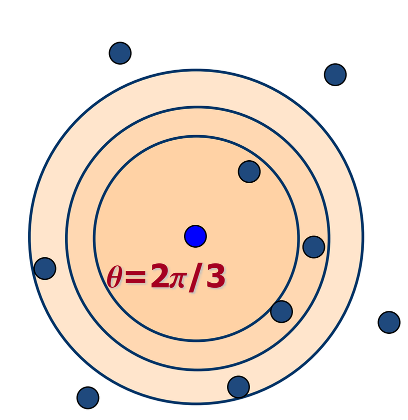
Choose optimal power levels from local geometric information to guarantee network connectivity in 3-D

Solution Approach: Algorithms

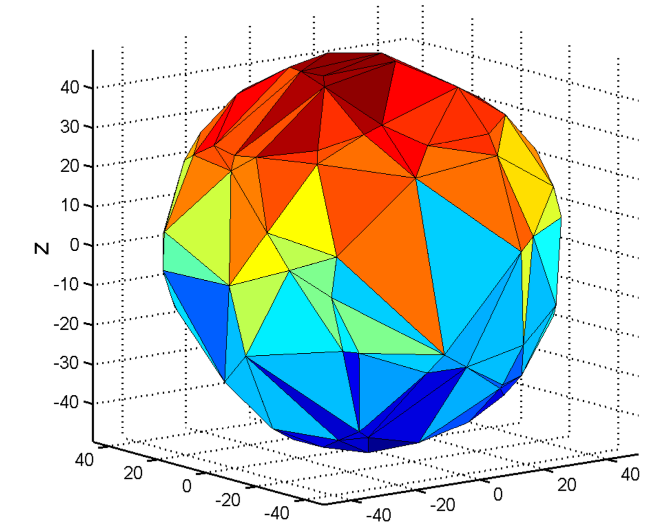
Orthographic Projection Based Approach

1. Uses multidimensional scaling to get relative location maps of nodes transmitting at max power
2. For a given power level (starting with min), each node projects its neighbors on the xy, yz, and zx plane
3. Checks if θ-constraint is satisfied on all the three planes; if not, go back to step 2, increment power level and repeat until max power; else, stop

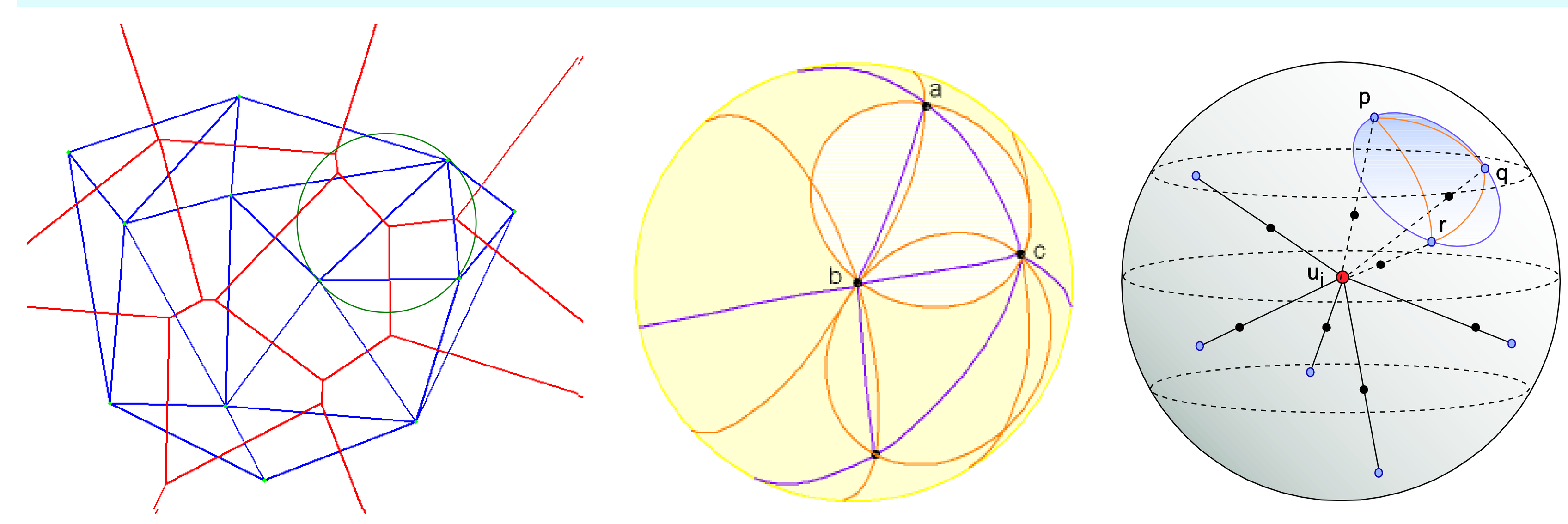




2D θ-constraint

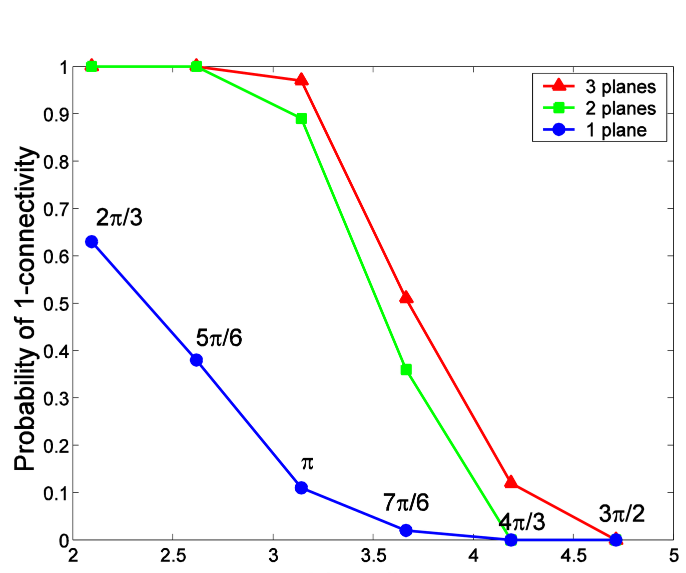


Spherical Delaunay Triangulation Based Approach



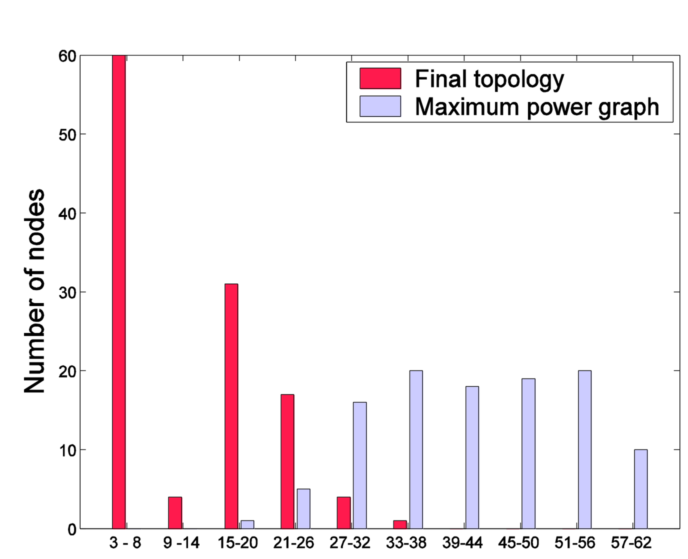
1. For a given power level (starting with min), each node projects its neighbors on the surface of its transmission sphere of radius R
2. Constructs Delaunay triangulation on the spherical surface and finds areas of empty caps (use of **Delaunay empty-circle property**)
3. If any cap area is > $2.7R^2$, go back to step 1, increment power level and repeat until max power; else, stop

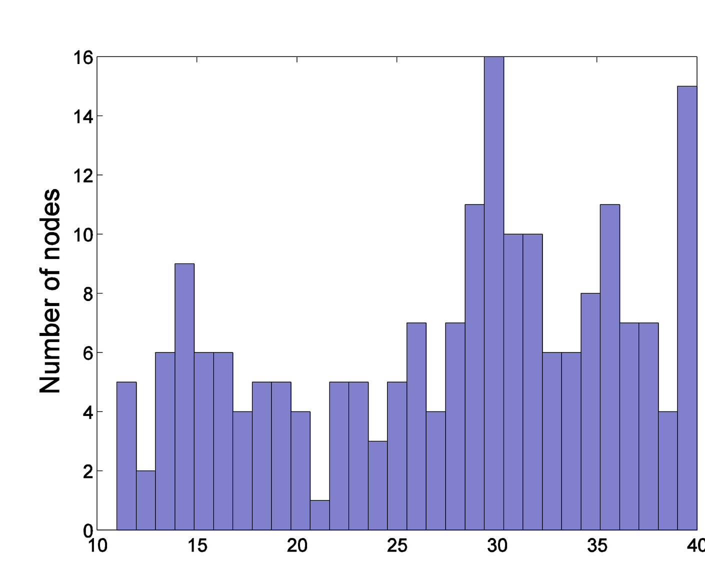
Performance Evaluation: Simulations

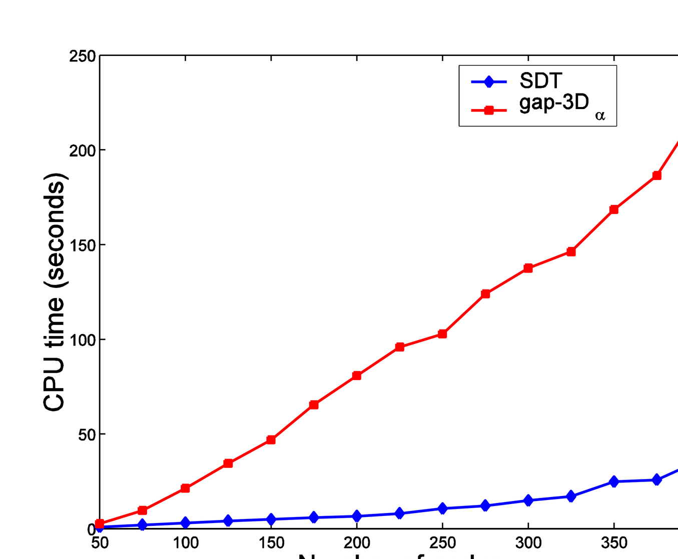


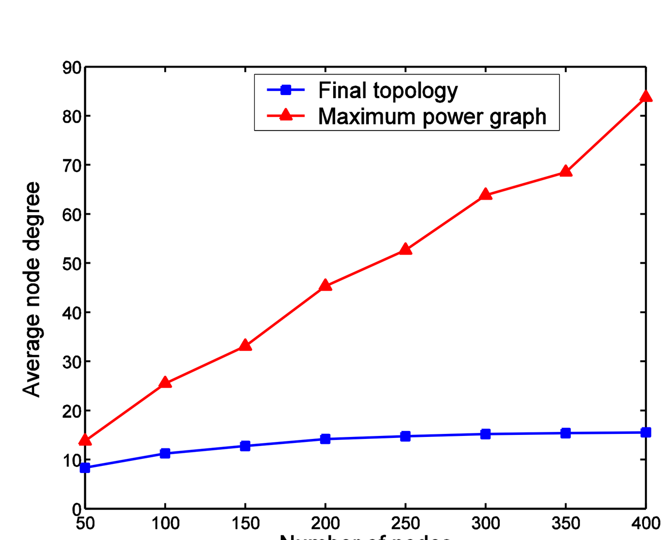
Orthographic projections: 100 nodes, connected network

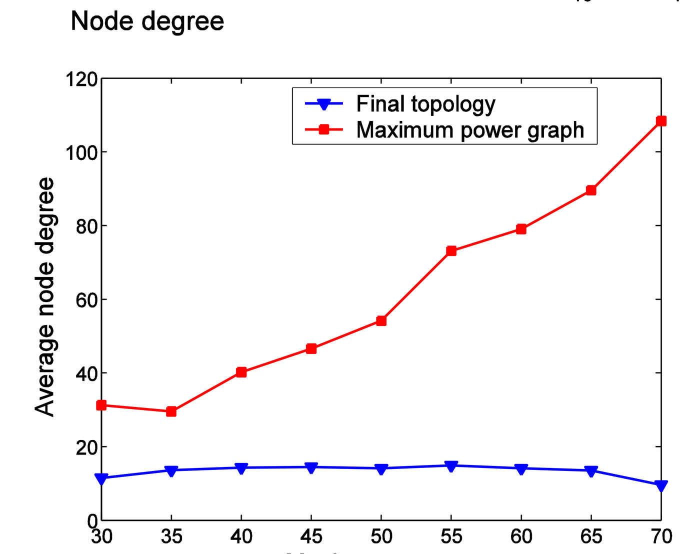
SDT simulation: 200 nodes, max power 40











Running time scales as O(d log d), an $O(d^2)$ improvement over prior art, d: average node degree

This work has been supported in part by NSF through grants number CNS-0435505, CNS-0347621, CNS-0627028, CCF-0430061 and ITR grant CNS-0325875